

PRACTICAL_NO_5

AIM: Probability and Probability Distributions:

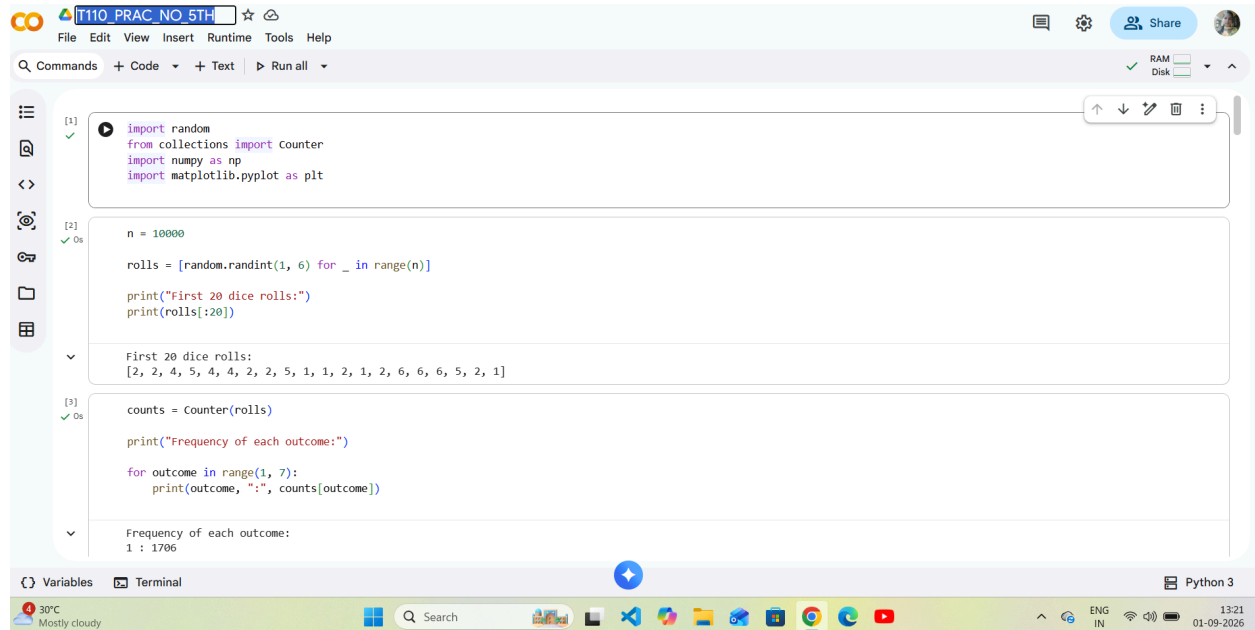
a. Simulate a probability experiment (e.g., rolling dice) using programming and calculate the probabilities of different outcomes.

b. Generate random numbers from various probability distributions (normal, uniform, exponential) and analyze their properties.

Import Libraries

Simulate Dice Rolling

Calculate Outcome Frequency



```
[1] import random
    from collections import Counter
    import numpy as np
    import matplotlib.pyplot as plt

[2] n = 10000

    rolls = [random.randint(1, 6) for _ in range(n)]

    print("First 20 dice rolls:")
    print(rolls[:20])

First 20 dice rolls:
[2, 2, 4, 5, 4, 2, 2, 5, 1, 1, 2, 1, 2, 6, 6, 6, 5, 2, 1]

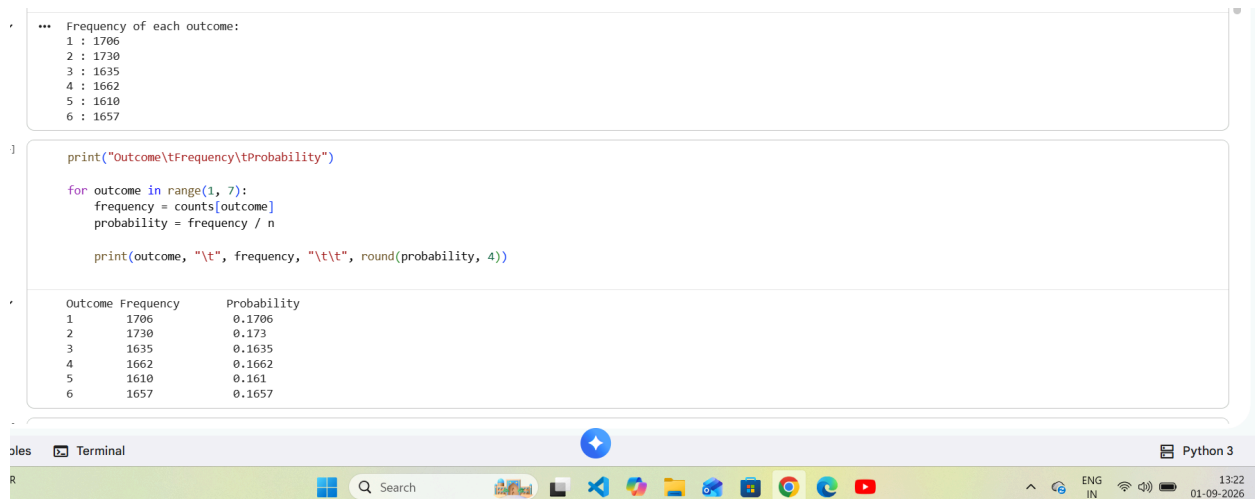
[3] counts = Counter(rolls)

    print("Frequency of each outcome:")

    for outcome in range(1, 7):
        print(outcome, ":", counts[outcome])

Frequency of each outcome:
1 : 1706
```

Calculate Empirical Probability



```
... Frequency of each outcome:
1 : 1706
2 : 1730
3 : 1635
4 : 1662
5 : 1610
6 : 1657

1 print("Outcome\tFrequency\tProbability")

    for outcome in range(1, 7):
        frequency = counts[outcome]
        probability = frequency / n

        print(outcome, "\t", frequency, "\t\t", round(probability, 4))

Outcome Frequency      Probability
1      1706      0.1706
2      1730      0.173
3      1635      0.1635
4      1662      0.1662
5      1610      0.161
6      1657      0.1657
```

Visualize Dice Probabilities

PRACTICAL_NO_5

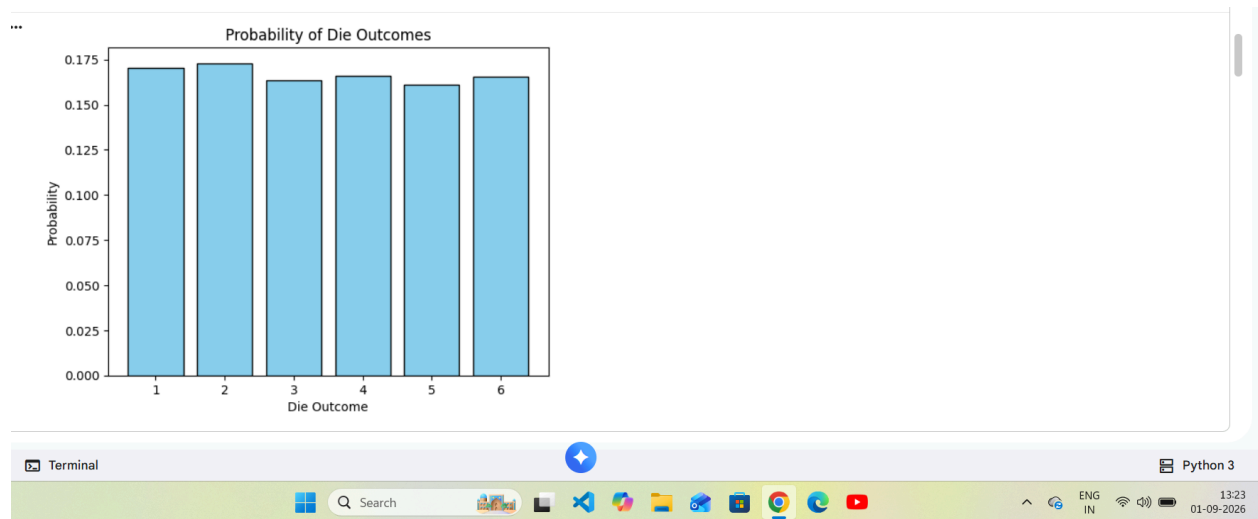
```
outcomes = list(range(1, 7))

probabilities = [counts[i] / n for i in outcomes]

plt.bar(outcomes, probabilities, color='skyblue', edgecolor='black')

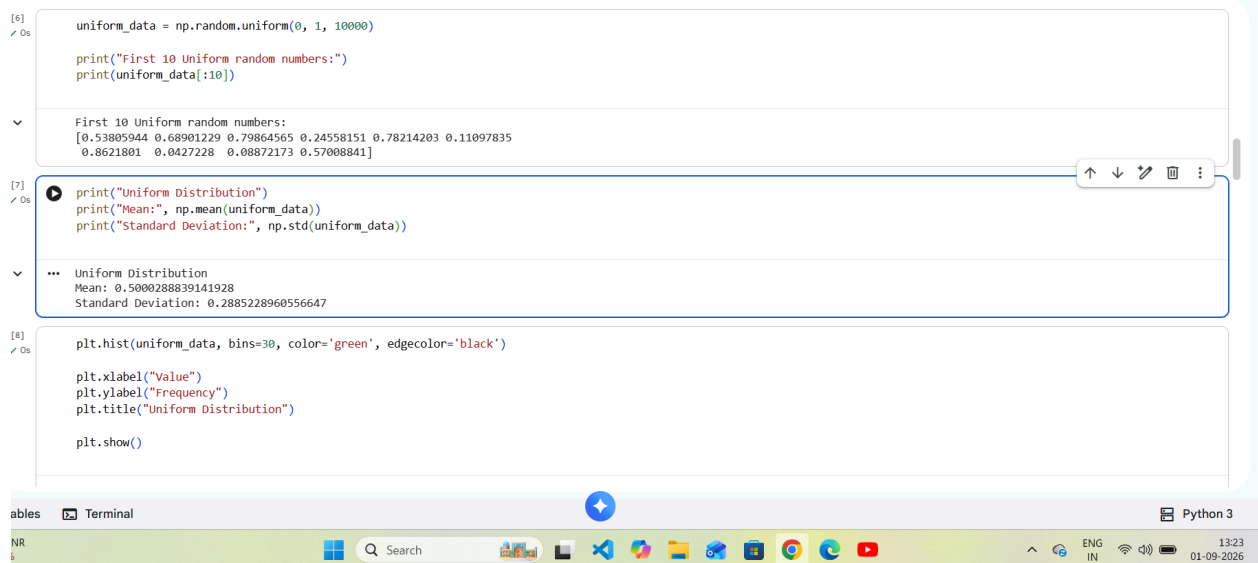
plt.xlabel("Die Outcome")
plt.ylabel("Probability")
plt.title("Probability of Die Outcomes")

plt.show()
```



Generate Uniform Distribution

Analyze Uniform Distribution



Generate Normal Distribution

TYCS_T110_SANJANA_SAHU

PRACTICAL_NO_5

Analyze Normal Distribution

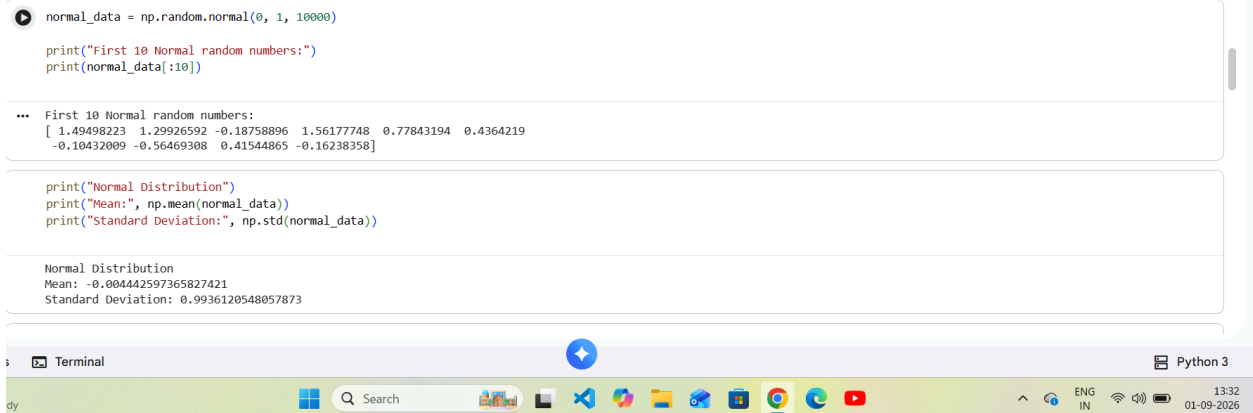
```
normal_data = np.random.normal(0, 1, 10000)

print("First 10 Normal random numbers:")
print(normal_data[:10])

... First 10 Normal random numbers:
[ 1.49498223  1.29926592 -0.18758896  1.56177748  0.77843194  0.4364219
 -0.10432009 -0.56469308  0.41544865 -0.16238358]

print("Normal Distribution")
print("Mean:", np.mean(normal_data))
print("Standard Deviation:", np.std(normal_data))

Normal Distribution
Mean: -0.004442597365827421
Standard Deviation: 0.9936120548057873
```



Generate Exponential Distribution

Analyze Exponential Distribution

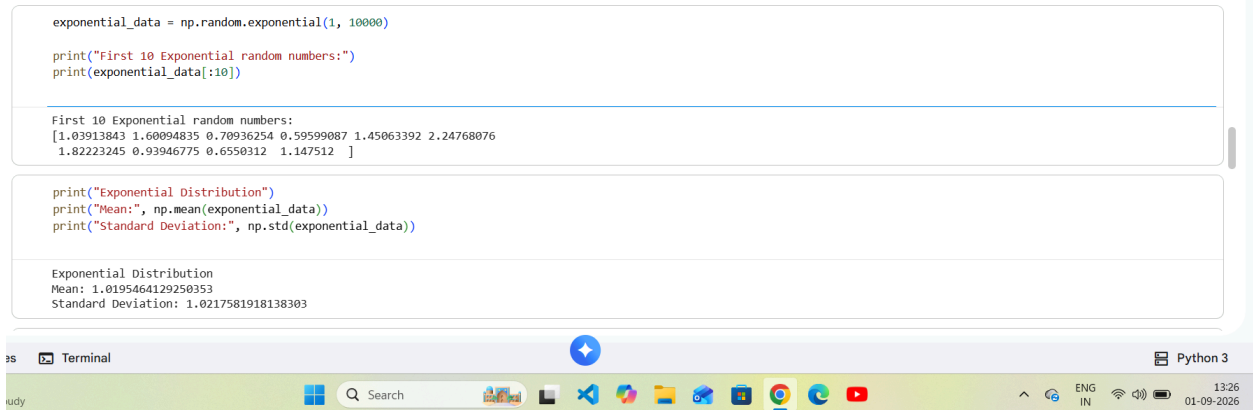
```
exponential_data = np.random.exponential(1, 10000)

print("First 10 Exponential random numbers:")
print(exponential_data[:10])

First 10 Exponential random numbers:
[1.03913843 1.60094835 0.70936254 0.59599087 1.45063392 2.24768076
 1.82223245 0.93946775 0.6550312 1.147512 ]

print("Exponential Distribution")
print("Mean:", np.mean(exponential_data))
print("Standard Deviation:", np.std(exponential_data))

Exponential Distribution
Mean: 1.0195464129250353
Standard Deviation: 1.0217581918138303
```



Visualize Uniform Distribution

Visualize Normal Distribution

Visualize Exponential Distribution

PRACTICAL_NO_5

```
[15]
✓ Os
plt.figure(figsize=(15, 4))

# Uniform Distribution
plt.subplot(1, 3, 1)
plt.hist(uniform_data, bins=30, color='green', edgecolor='black')
plt.title("Uniform Distribution")
plt.xlabel("Value")
plt.ylabel("Frequency")

# Normal Distribution
plt.subplot(1, 3, 2)
plt.hist(normal_data, bins=30, color='orange', edgecolor='black')
plt.title("Normal Distribution")
plt.xlabel("Value")
plt.ylabel("Frequency")

# Exponential Distribution
plt.subplot(1, 3, 3)
plt.hist(exponential_data, bins=30, color='red', edgecolor='black')
plt.title("Exponential Distribution")
plt.xlabel("Value")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```

Variables Terminal Python 3

30°C Mostly cloudy 13:39 01-09-2026

